A SURJECTIVE PROPER MORPHISM OF ALGEBRAIC SPACES IS A PH COVERING

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Weak Chow's lemma for algebraic spaces. Cohomology of Spaces, lemma weak-chow (cross-chapter, named). Produces the surjective proper morphism of schemes V -> U factoring through Y x_X U.

Source reference. Original reference 089J: focused-seeds/geometry/S9-topo-surjective-proper-ph/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a surjective proper morphism of algebraic spaces over S. Then $\{Y \to X\}$ is a ph covering.*

Proof. Let $U \to X$ be a morphism with U affine. By Chow's lemma (in the weak form given as Cohomology of Spaces, Lemma 089J) we see that there is a surjective proper morphism of schemes $V \to U$ which factors through $Y \times_X U \to U$. Taking any finite affine open cover of V we obtain a standard ph covering of U refining $\{X \times_Y U \to U\}$ as desired. □

Date: stacks-project:spaces-topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.